Net force first

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Combine forces before using mass to predict acceleration.

SUCCESS CRITERIA - I can...

1. Find net force before dividing by mass.
2. State acceleration with units and direction.
3. Separate acceleration direction from motion direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “mass” to explain your answer. Sentence starters are optional.

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A **5 kg** sign hangs still from one vertical cord.
1. Use g=**10 N/kg**.
2. **Name both forces**, their sources, magnitudes and directions.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: ΣF = ma; a = ΣF/m.
Add signed forces on the same object before dividing by mass. N/kg = m/s².

Worked reasoning: A **2 kg** cart has **9 N** right and **3 N** left. Net force=**6 N** right; a=**6/2**=**3 m/s²** right. **Add forces before dividing** by mass.

Your turn: 1. **Draw arrows** for both forces, then a separate net-force arrow.
2. **Explain why 9/2** is not the acceleration.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



2 kg cart: draw the net force separately.

Marzano check - circle: ① ② ③

03 Read the arrows

For the **4 kg** cart, find net horizontal force and acceleration. **Write each step**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



The dot represents the object. Mass: 4 kg

Marzano check - circle: ① ② ③

04 Balanced but moving

A **2 kg** cart is already moving left when two forces act: **5 N** right and **5 N** left.
1. **Use the diagram** to **find its acceleration**.
2. **Describe its motion** if these forces remain balanced.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Compare two masses

Two test carts experience the same **6 N** net force. **Complete the comparison table**. Which cart changes velocity faster, and what causes the difference?

Sentence starter: Both … . They differ in … because … .

Mass (kg)	Net force (N)	a (m/s²)
2	6	____
3	6	____

Marzano check - circle: ① ② ③

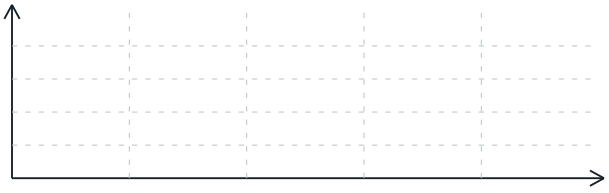
06 Review: connect an earlier idea

A cart is at **5**, **12**, **19 m** at **0**, **1**, **2 s**. **Predict its position** at **9 s**. **State something you** are treating as true and a measurement that could challenge it.
Label the diagram with the given positions and any prediction.

Word help: prediction = what a model says may happen next

Sentence starter: I predict … . I used … to make this prediction because … .

Position (m)



Time (s)

Marzano check - circle: ① ② ③

07 Exit check

An **8 kg** object has **50 N** right and **18 N** left. **Draw the forces**, calculate net force, and **find a** with direction.

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Draw the given forces; show net force separately.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. **Find two different pairs of** opposite horizontal forces that give the same **6 N** net force right.
2. Keep mass **2 kg** and **compare accelerations**.

Sentence starter: Both … . They differ in … because … .

Add and label force arrows.

Label each interaction force on the cart.

Marzano check - circle: ① ② ③